

AWS Identity and Access Management (IAM)

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Security is Job Zero at Amazon

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...because getting started requires knowledge of IAM

and things are not obvious

Policies and Permissions

Example: Who can read / write to this S3 bucket?

Policies and Permissions provide this access

A **policy** is an object in AWS that, when associated with an identity or resource, defines permissions

Create policies and attach to IAM identities (users, groups of users, or roles) or AWS resources

Users

We create IAM users in an AWS account

May be human user or workload interacting with AWS resources

IAM user consists of name and credentials

IAM User Groups

An IAM user group is a collection of IAM users

Lets you specify permissions for multiple users

Can make it easier to manage permissions for those users

Roles

As a data science manager, I want all of my data scientists to have the same permissions.

Do I need to set the same permissions for each user?

No, you can create an **IAM role**

IAM role is an identity you can create with specific permissions e.g., the *data_scientist* role

Resources

For more details, see the documentation:

<https://docs.aws.amazon.com/IAM/latest/UserGuide/introduction.html>